

WEEK 1 ASSIGNMENT

AI Foundations & Developer Environment

Name: Gundavaram Aswini

Roll Number: 24JR1A0510

1. What is Artificial Intelligence?

Artificial Intelligence, also called AI, is a technology that helps computers and machines think and work like humans. AI can learn from data, solve problems, answer questions, and make decisions. Today AI is used in many applications like Google Assistant, Alexa, chatbots, YouTube recommendations, self-driving cars, and online shopping apps. AI makes work easier and faster. It can do repetitive tasks without getting tired. In schools and colleges, students use AI tools for learning, coding, and project ideas. Many companies use AI to improve customer service and save time. AI is becoming very important in daily life and in the future it will be used even more in every field like healthcare, education, banking, transportation, and business.

2. Difference between AI, Machine Learning, Generative AI, and Agentic AI

Artificial Intelligence is the main technology that allows machines to act smart like humans. Machine Learning is a part of AI where computers learn from data automatically. Instead of writing every instruction, the machine learns patterns from examples. Generative AI is another type of AI that can create new content such as text, images, music, videos, and code. ChatGPT is a popular example of Generative AI because it can answer questions and generate content. Agentic AI is a more advanced AI system that can make decisions and complete tasks by itself. It can plan actions and solve problems with less human help.

3. Explain any three open-source AI models available on Hugging Face

Hugging Face is a popular platform where developers can use and share AI models. Many open-source AI models are available on this platform. 1. Llama 3 – Used for chatting, answering questions, coding help, and content generation. 2. Mistral 7B – A lightweight and powerful AI model useful for AI projects and research. 3. Stable Diffusion – An AI model used for image generation from text prompts. These open-source models help students and developers learn AI and build projects easily.

4. Difference between traditional programming and AI-assisted development

Traditional programming means developers write all the code and instructions manually. The computer only follows the rules written by the programmer. AI-assisted development means developers use AI tools to help in coding. AI can suggest code, find errors, explain concepts, and improve productivity. This makes development faster and easier.

5. Explain AI hallucinations with one example

AI hallucination means an AI system gives wrong or fake information confidently. Sometimes AI creates answers that look correct but are actually false. For example, if an AI chatbot says that “Albert Einstein invented the computer,” it is incorrect information. This is called an AI hallucination. Users should always verify important information from trusted sources.

Conclusion

Artificial Intelligence is changing the world and making technology smarter. Technologies like Machine Learning, Generative AI, and Agentic AI are helping people in many industries. Learning AI concepts is important for students because AI will play a major role in the future.

